Consistency of Compressible Momentum Thickness

A short proof concerning Schlichting's Eqs. (10.92) and (10.96)

Technical note

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Scope. Schlichting's momentum-integral equation, Eq. (10.92), is accepted as given, including the assumptions under which it was derived. The purpose is only to determine which profile integral can represent the momentum thickness in that equation. A fixed upper limit h is taken in uniform edge flow; it may equally be replaced by the boundary-layer edge or by infinity.

Proposition 1. *With the standard zero-defect normalization, the momentum thickness compatible with Schlichting's Eq. (10.92) is*

$$\boxed{\delta_2 = \theta_{\mathrm{c}} = \int_0^h \frac{\rho u}{\rho_e u_e}\left(1 - \frac{u}{u_e}\right)\mathrm{d}y.} \tag{1}$$

The expression printed as Schlichting's Eq. (10.96), if read literally, is not generally compatible with Eq. (10.92) unless an additional density-defect term is retained.

Proof. Schlichting's Eq. (10.92) is

$$\delta_2' + \frac{\delta_2}{u_e}u_e'\left(2 + \frac{\delta_1}{\delta_2} - M_e^2\right) = \frac{\tau_w}{\rho_e u_e^2}. \tag{S10.92}$$

The edge relation

$$\frac{\rho_e'}{\rho_e} = -M_e^2\frac{u_e'}{u_e} \tag{2}$$

is already implicit in the $-M_e^2$ term. Multiplication of Eq. (S10.92) by $\rho_e u_e^2$, followed by the product rule and Eq. (2), gives

$$\boxed{\tau_w = \frac{\mathrm{d}}{\mathrm{d}x}(\rho_e u_e^2\delta_2) + \rho_e u_e \delta_1 u_e'.} \tag{3}$$

It remains to identify the first term physically. Define

$$I_m = \int_0^h \rho u\,\mathrm{d}y, \qquad I_p = \int_0^h \rho u^2\,\mathrm{d}y.$$

Integration of steady mass and streamwise momentum conservation through the boundary layer gives

$$I_m' + \rho_e v_e = 0, \qquad I_p' + \rho_e u_e v_e = -h p_e' - \tau_w. \tag{4}$$

Using the inviscid edge relation $p_e' = -\rho_e u_e u_e'$ to eliminate p_e', and using the first of Eqs. (4) to eliminate v_e, yields

$$\tau_w = \frac{\mathrm{d}}{\mathrm{d}x}(u_e I_m - I_p) + u_e'(\rho_e u_e h - I_m). \tag{5}$$

Now introduce Schlichting's displacement thickness, Eq. (10.95),

$$\delta_1 = \int_0^h \left(1 - \frac{\rho u}{\rho_e u_e}\right) \mathrm{d}y, \tag{6}$$

and note that

$$u_e I_m - I_p = \int_0^h \rho u(u_e - u)\, \mathrm{d}y \equiv J. \tag{7}$$

Equation (5) therefore becomes the physical momentum-deficit balance

$$\boxed{\tau_w = J' + \rho_e u_e \delta_1 u_e'.} \tag{8}$$

Equations (3) and (8) contain identical wall-stress and displacement-thickness terms. Hence

$$\frac{\mathrm{d}}{\mathrm{d}x}(\rho_e u_e^2 \delta_2 - J) = 0. \tag{9}$$

The standard normalization that both quantities vanish with the momentum deficit removes the possible x-independent constant. Thus

$$\rho_e u_e^2 \delta_2 = J = \int_0^h \rho u(u_e - u)\, \mathrm{d}y,$$

which, after division by $\rho_e u_e^2$, proves Eq. (1).

For comparison, a literal reading of Schlichting's Eq. (10.96) defines

$$\theta_\mathrm{S} = \int_0^h \frac{\rho u}{\rho_e u_e}\left(1 - \frac{\rho u}{\rho_e u_e}\right) \mathrm{d}y. \tag{S10.96}$$

Multiplying the two candidate definitions by $\rho_e u_e^2$ gives

$$\rho_e u_e^2 \theta_\mathrm{c} = \int_0^h (\rho u u_e - \rho u^2)\, \mathrm{d}y, \tag{10}$$

$$\rho_e u_e^2 \theta_\mathrm{S} = \int_0^h \left(\rho u u_e - \frac{\rho^2 u^2}{\rho_e}\right) \mathrm{d}y. \tag{11}$$

The momentum equation transports ρu^2, as present in Eq. (10), not $\rho^2 u^2/\rho_e$ as present in Eq. (11). Therefore Eq. (S10.96) is not the momentum-deficit coefficient in Eq. (8) for a general variable-density profile. □

Exact consequence of using Eq. (S10.96). Let

$$\Delta\theta \equiv \theta_\mathrm{S} - \theta_\mathrm{c} = \int_0^h \frac{\rho u}{\rho_e u_e}\frac{u}{u_e}\left(1 - \frac{\rho}{\rho_e}\right) \mathrm{d}y. \tag{12}$$

Since $\theta_\mathrm{c} = \theta_\mathrm{S} - \Delta\theta$, the correct balance becomes

$$\tau_w = \frac{\mathrm{d}}{\mathrm{d}x}(\rho_e u_e^2 \theta_\mathrm{S}) + \rho_e u_e \delta_1 u_e' - \frac{\mathrm{d}}{\mathrm{d}x}(\rho_e u_e^2 \Delta\theta). \tag{13}$$

Equivalently, Eq. (10.92) written with θ_S requires the residual

$$\boxed{\mathcal{R}_S = \frac{1}{\rho_e u_e^2}\frac{\mathrm{d}}{\mathrm{d}x}(\rho_e u_e^2 \Delta\theta) = \Delta\theta' + (2 - M_e^2)\frac{u_e'}{u_e}\Delta\theta.} \tag{14}$$

Constant density is a sufficient special case for $\Delta\theta = 0$; it is not an assumption made in the proof.

Assumptions used. Equation (S10.92) and its original assumptions are accepted. The short derivation of Eq. (8) uses the same standard ingredients: steady two-dimensional thin-boundary-layer conservation, a stationary impermeable wall, a uniform edge state, no streamwise body force, and the inviscid edge momentum relation. Equation (2), already encoded in Eq. (S10.92), is the additional isentropic edge relation needed for its Mach-number form. No constant-density, zero-pressure-gradient, or thermal-wall assumption is introduced.

Cross-check with Schlichting. Equation (6) is exactly Schlichting's Eq. (10.95), while Eq. (S10.92) is his Eq. (10.92). His Eqs. (10.97) and (10.98) also retain the expected pattern of a mass-flux weight multiplied by a deficit in the transported quantity. Independent standard references print Eq. (1): White and Majdalani, Eq. (7-75), and Anderson, Eq. (6.137) [2, 3]. These checks localize the inconsistency to the second density factor in the printed Eq. (10.96). The argument proves incompatibility of the two printed equations; whether the cause is a typographical error is an editorial question.

References

[1] H. Schlichting and K. Gersten, *Boundary-Layer Theory*, 9th ed., Springer, 2017, Eqs. (10.92)–(10.98).

[2] F. M. White and J. Majdalani, *Viscous Fluid Flow*, 4th ed., McGraw Hill, 2022, Sec. 7-4.2, Eq. (7-75), p. 418.

[3] J. D. Anderson, Jr., *Hypersonic and High-Temperature Gas Dynamics*, 2nd ed., AIAA, 2006, Sec. 6.7.7, Eq. (6.137), p. 333.